CTFVault - Relational Schema

Underlined = primary key. Blue = foreign key. Arrows point from each foreign key to the primary key it references.

CATEGORY

category_id	name

TEAM

team_id	name	invite_code	created_at

UNIQUE(name) UNIQUE(invite_code)

USER

user_id	username	email	password_hash	role	team_id	is_captain	created_at

ON DELETE SET NULL CHECK role IN ('admin','setter','player') UNIQUE(username) UNIQUE(email)

CHALLENGE

challenge_id	title	description	category_id	author_id	initial_points	min_points	decay_rate	current_points	solve_count	max_solves	is_locked	released_at

CHECK min_points <= initial_points UNIQUE(title)

CHALLENGE_FLAG

flag_id	challenge_id	team_id	flag_hash	generated_at

UNIQUE(challenge_id, team_id)

SUBMISSION

submission_id	challenge_id	team_id	user_id	submitted_hash	is_correct	is_first_blood	submitted_at

UNIQUE(challenge_id, team_id) WHERE is_correct UNIQUE(challenge_id) WHERE is_first_blood

SCORE_LEDGER

entry_id	team_id	challenge_id	submission_id	points	entry_type	awarded_at

CHECK entry_type IN ('solve','first_blood','decay_adjust','hint_penalty')

WRITEUP

writeup_id	challenge_id	team_id	author_id	title	body	search_vector	is_published	created_at

UNIQUE(challenge_id, team_id)

WRITEUP_CHUNK

chunk_id	writeup_id	chunk_index	content	embedding

ON DELETE CASCADE UNIQUE(writeup_id, chunk_index)

HINT

hint_id	challenge_id	content	cost	order_index	embedding

ON DELETE CASCADE UNIQUE(challenge_id, order_index)

HINT_UNLOCK

team_id	hint_id	cost_paid	unlocked_at

COLLUSION_ALERT

alert_id	writeup_a_id	writeup_b_id	distance	status	reviewed_by	detected_at	resolved_at

CHECK writeup_a_id < writeup_b_id UNIQUE(writeup_a_id, writeup_b_id)

AUDIT_LOG

log_id	actor_id	table_name	record_id	action	old_value	new_value	occurred_at